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Seroprevalence and incidence of transfusion-transmitted infectious diseases among blood donors from regional blood transfusion centres in Burkina Faso, West Africa

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Abstract

BACKGROUND AND OBJECTIVE The high prevalence of numerous transfusion-transmitted infectious diseases such as HIV, HBV, HCV and syphilis in sub-Saharan Africa affects blood safety for transfusion recipients. The aim of this study was to evaluate the prevalence and incidence of transfusion-transmissible infectious diseases among blood donors in Burkina Faso.

METHODS A retrospective study of blood donors' records from January to December 2009 was conducted. Prevalence and incidence of viral infections were calculated among repeat and first-time blood donors.

RESULTS Of the total of 31 405 first-time volunteer blood donors in 2009, 24.0% were infected with at least one pathogen and 1.8% had serological evidence of multiple infections. The seroprevalence of HIV, HBV, HCV and syphilis in first-time volunteer donors was 1.8%, 13.4%, 6.3% and 2.1%, respectively. In 3981 repeat donors, the incidence rate was 3270.2, 5874.1 and 6784.6 per 100 000 donations for anti-HIV-1, HBsAg and anti-HCV, respectively. These numbers varied significantly according to populations where blood is collected and blood centres in Burkina Faso.

CONCLUSION The relatively high prevalence of viral markers in first-time volunteers and remarkably high incidence of infections in repeat donors raise concerns regarding the safety of these donors and suggest that implementation of NAT might significantly improve the situation.

keywords HIV, HBV, HCV, transfusion, blood donors

Introduction

Blood safety remains a major public health problem in sub-Saharan Africa, because of high prevalence of infections and lack of financial resources responsible for inappropriate infrastructures and under-qualified personnel (Tessema *et al.* 2010). There are different systems providing for the blood supply in sub-Saharan Africa. The National Blood Transfusion Centre of Burkina Faso (CNTS) has opted for a centralized system that recruits non-remunerated voluntary donors (VNRD). CNTS includes four regional blood transfusion centres: based at Ouagadougou, Bobo-Dioulasso, Koudougou and Fada N'gourma. However, in 2008, 40% of the blood was collected in other institutions, mostly hospitals relying mostly on family/replacement donors (Dahourou *et al.* 2010).

Screening for transfusion-transmissible infections such as HIV, HBV, HCV and syphilis is essential to insure blood transfusion safety and by extension for the protection of human life. For many years, VNRD were considered safer than family donors. This widely disseminated belief was not supported by evidence in four sub-Saharan African countries surrounding Burkina Faso (Allain *et al.* 2010; Loua & Nze Nkoure 2010; Mbanya *et al.* 2010; Diarra *et al.* 2009). The assumed safety benefit of VNRD was in fact related to a minority of donors who repeated donation, artificially decreasing marker prevalence. For the lack of computerized record of donors and donations, very few data were available to calculate incidence of viral infections. Such system being now available to the Burkina Faso national blood services; the first year of data is reported here.

Blood transfusion was considered to account for 5–10% of HIV infections in sub-Saharan Africa (UNAIDS 2002;

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Moore *et al.* 2001), and 12.5% of patients who received blood transfusion are at risk of post-transfusion hepatitis (Fasola & Otegbayo 2002). In Burkina Faso, the prevalence of HBV and HCV in the general population is high with wide local variations (Collenberg *et al.* 2006). In 2009, the prevalence of HIV in the general population was estimated at 1.2% (UNAIDS 2010, http://www.unaids.org/documents/20101123_2010_HIV_Prevalence_Map_em.pdf), and syphilis seroprevalence among blood donors was reported at 1.6% (Kania *et al.* 2009). This study was undertaken to estimate both prevalence and incidence of the main infectious agents in our volunteer donor population and to determine the potential need of further blood safety measures and options to reduce the blood shortage.

Materials and methods

Donors

A retrospective analysis of donor data from January to December 2009 in three regional blood transfusion centres (Ouagadougou, Fada N'gourma and Bobo-Dioulasso) was conducted. Voluntary donors were all healthy subjects, selected after responding to a panel of questions comprising a medical history. Apparently healthy individuals aged 17–65 years with a weight >50 kg were eligible for blood donation. All donors answered questions intending to exclude recipients of previous transfusion, individuals having experienced jaundice or signs of hepatitis, pregnant women and people having experienced high-risk sexual behaviour within 2 weeks preceding the intended donation. Socio-demographic characteristics of selected donors were recorded in a database, and venous blood was collected in blood banking bags following standard procedures.

In each centre, blood collection was conducted according to two main systems. Approximately 2/3 of the blood was collected at the regional blood centres in the three main cities. Donors were mostly adult urban dwellers. The last third of the donations was collected in mobile sessions set essentially in secondary schools but also in universities, places of worship such as churches or mosques, barracks, community associations, large businesses and at socio-cultural events where a large number of people were expected to assemble. In addition, some blood collection sessions were taking place outside the cities in village town halls or secondary schools (districts). The distribution of donors between these various sources varied according to location.

Ethical considerations

This study was approved by the CERBA/Saint Camille Ethics Committee. However, because of the retrospective nature of the study, informed consent was not obtained from the study subjects.

Serological analysis

Hepatitis B surface antigen (HBsAg), antibodies to *Treponema pallidum* and HCV were detected using Hepanostika HBsAg Ultra (Biomérieux, Boxtel, the Netherlands), Rapid Plasma Reagin test (RPR; Cypress Diagnostics, Langdorp, Belgium) and Hepanostika HCV Ultra (Beijing United Biomedical Co. Ltd., Beijing, China), respectively. Antibodies to HIV types 1 and 2 were screened for using Vironostika HIV Uni-Form II Ag/Ab (Biomérieux, Boxtel, the Netherlands). All samples reactive for HIV, HBsAg and HCV were retested for confirmation using a second enzyme-linked immunosorbent assay (Bio-Rad, Marnes la Coquette, France). The presence of antibodies to *T. pallidum* was confirmed with a *T. pallidum* haemagglutination test (TPHA; Cypress Diagnostics, Langdorp, Belgium). A result was considered positive if both the first and second tests were positive.

Statistical analysis

Data were analysed using EPI-Info version 6.04 dfr (CDC, Atlanta, GA, USA). Odds ratio was calculated to determine risk factors associated with HIV, HBV, HCV and syphilis. *P* values below 0.05 were considered statistically significant.

Incidence rates

Incidence rates (IR) for repeat donors were calculated according to Schreiber *et al.* (1996), Busch *et al.* (2005) and O'Brien *et al.* (2007) by dividing the number of incidence cases in the study period by the total number of person-years. Person-years were calculated by adding the inter-donation intervals for all donors but only taking half the interval for incidence cases, assuming that the infection was acquired half way between the last negative and the first positive donation. IRs for first-time donors were calculated by multiplying the IR/100 000 donations in repeat donors by a conversion factor. The conversion factor was estimated from the data as the IRs in Burkina Faso are much higher that those in other countries, where a conversion factor of between 2.4 and 3.2 was estimated for HIV and HCV (Janssen *et al.* 1998; Dodd *et al.* 2002; Stramer *et al.* 2004). The conversion factor was calculated by dividing the number of positive donations by the total number of donations separately for first-time and repeat donors and then dividing this result for first-time donors by that for repeat donors.

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Results

Demographic characteristics of blood donors

From a total of 35 401 blood donors, 56.6% were in the age group of 18–25 years, and the median age of study subjects was 24 years (range 17–67 years). Of these, 74.9% donors were men and 25.1% were women. There were 31 405 first-time donors (88.7%) and 3996 repeat donors (11.3%). The percentage of repeat donors was 11.3% ranging between 7.9% and 12.8% according to centres. Among the repeat donors, 70.4% gave two donations, 22.4% three donations and 7.2% four times (Table 1). Blood donors were mainly recruited from schools (31.8%), blood transfusion centres (27.7%) and districts (10.4%).

Prevalence of infectious agent markers in first-time volunteer donors

The prevalence of infection markers in first-time volunteer donors is shown in Table 2. The details of single-marker and co-infections are given for the total population and for each blood centre. In the overall donor population, the prevalence of anti-HIV, HBsAg, anti-HCV and anti-treponema was 1.8%, 13.4%, 6.3% and 2.1%, respectively. Each marker had a clearly lower prevalence in the Bobo-Dioulasso blood centre ($P < 0.001$). In contrast, while Ouagadougou population has a significantly higher prevalence of anti-HIV, Fada'Ngourma donors carry a significantly higher prevalence of both HBsAg and anti-HCV. The differences in prevalence of all four markers were significant with a P value <0.001 between centres except for HIV between Bobo-Dioulasso and Fada'N-gourma. The percentage of first-time volunteer blood being discarded was 17% in Bobo-Dioulasso but 23.1% and 23.9% in Ouagadougou and Fada'Ngourma, respectively.

Overall, 1.7% (589/35 401) of donors were co-infected. In a population with high prevalence of chronic HBV infection, HIV and HBsAg were associated in line with the prevalence (15.9%). The rate of association of anti-HCV with either anti-HIV or HBsAg was not similar and discordant with the overall prevalence (0.1% and 0.9%, respectively). However, in the Fada N'gourma blood centre, HBsAg/anti-HCV association was more frequent than in the other blood centres.

Incidence of infectious agent markers in repeat volunteer donors

A total of 3996 donors repeated donating 2–4 times in 2009. Calculating the intervals between these 8384 donations and using the incidence of seroconversion to anti-HIV, HBsAg or anti-HCV, the IRs of each virus were calculated (Table 3). Similar to what was observed in first-time donors' viral marker prevalence, there was considerable differences between centres, Bobo-Dioulasso population of repeat donors showing a lower IR than that in the other two centres. However, overall, remarkably high IRs were observed. When using the alternative method to calculate IRs, based on prevalence data in first-time volunteer donors, considerable differences were seen, the IR being approximately three times higher for HIV, 10 times for HBV infection and five times for HCV infection (Table 3).

Incident infections occurred essentially in men (88.5%), irrespective of the marker (range 86.9–90.2%). In contrast, while repeat donors came essentially from three types – blood centres (66.8%), schools (21.3%) and districts (5.8%) – HBV and HCV infections originated disproportionately from district donors (22.2% and 19.1%, respectively) and to a lesser extent from secondary school students (37.4% and 26.4%, respectively). Overall, 5.6% repeat blood donors attending blood centres

Table 1 Percentage of first-time and repeat blood donors among blood transfusion centres in Burkina Faso in 2009

Characteristics	Ouagadougou (%)	Bobo-Dioulasso (%)	Fada N'gourma (%)	Total (%)
Total number of donors	19 582	9434	6385	35 401
N first-time donors	17 294	8231	5880	31 405
N repeat donors†	2288 (11.7)	1203 (12.8)	505 (7.9)	3996 (11.3)
Two donations‡	1642 (71.8)	782 (65.0)	388 (76.8)	2812 (70.4)
Three donations	501 (21.9)	290 (24.2)	105 (20.8)	896 (22.4)
Four donations	145 (6.3)	131 (10.8)	12 (2.4)	288 (7.2)
Male	1815 (79.3)	1024 (85.1)	391 (77.4)	3230 (80.8)
Female	473 (20.7)	179 (14.9)	114 (22.6)	766 (9.2)

†The percentage is from total number of donors in each blood transfusion centre.
‡The percentages given below represent percentages from total number of repeat donors.

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Table 2 Prevalence of confirmed single-marker infections as well as co-infections in first-time volunteer blood donors

Co-infections	Anti-HIV	HBsAg	Anti-HCV	Anti-syphilis	Other combinations†	Total (%)
Anti-HIV	**433**	77	35	11	IBCS 1, IBC 6,	567 (1.8)
HBsAg		**3706**	298	101	IBS 2, ICS 2,	4203 (13.4)
Anti-HCV			**1566**	44	BCS 12	1964 (6.3)
Anti-syphilis				**491**		664 (2.1)
Ouagadougou						
Anti-HIV	**321**	56	28	10	IBCS 1, IBC 5,	425 (2.5)
HBsAg		**2028**	158	75	IBS 2, ICS 2,	2334 (13.5)
Anti-HCV			**886**	27	BCS 9	1116 (6.5)
Anti-syphilis				**371**		497 (2.9)
P value O vs. B	<0.0001	<0.0001	<0.0001	<0.0001		
OR (95% IC)	2.5 (2.0–3.2)	1.3(1.2–1.4)	1.5 (1.4–1.7)	5.0 (3.7–6.9)		
Bobo-Dioulasso						
Anti-HIV	**70**	8	3	0	BCS 2	81 (1.0)
HBsAg		**839**	43	9		901 (11.0)
Anti-HCV			**300**	5		353 (4.3)
Anti-syphilis				**32**		48 (0.6)
P value B vs. F	0.75	<0.0001	<0.0001	<0.0001		
OR (95% IC)	1.1 (0.8–1.5)	1.6(1.4–1.8)	1.9 (1.6–2.2)	3.5 (2.5–5.0)		
Fada N'gourma						
Anti-HIV	**42**	13	4	1	IBC 1, BCS 1	61 (1.1)
HBsAg		**839**	97	17		968 (18.0)
Anti-HCV			**380**	12		495 (9.2)
Anti-syphilis				**88**		119 (2.2)
P value O vs. F	<0.0001	<0.0001	0.0003	0.0004		
OR (95% IC)	2.4 (1.8–3.2)	1.3(1.2–1.4)	1.2 (1.1–1.4)	1.4 (1.2–1.8)		

O, Ouagadougou; B, Bobo-Dioulasso; F, Fada N'gourma; OR, Odds ratio; IC, Confidence Interval.
The numbers in bold indicate the number of single-marker infections in each population. Other numbers indicate dual-marker infections.
†Samples with more than two positive markers are detailed in the sixth column where I = anti-HIV, B = HBsAg, C = anti-HCV, S = anti-treponema, followed by the number of samples affected.

seroconverted to a viral marker. Donors donating at other types of circumstances such as in schools and in districts had a significantly higher rate of seroconversion (9.5% and 22.8%, respectively; $P < 0.0001$ between blood centres and schools or districts and $P = 0.0032$ between schools and districts). HIV infections were evenly distributed according to donors' age. HBV and HCV infections were significantly more frequent in donors below age 20 who represented 15.3% of repeat donors but accounted for 27.6% and 22.7% of HBV and HCV infections.

Discussion

In recent years, the issue of HIV blood safety has been in the forefront in sub-Saharan African countries, leaving HBV and HCV safety far behind exemplified by a considerably lower percentage of blood units tested for the two latter markers than for the former. One highly recommended approach was to recruit voluntary non-remunerated donors and to exclude family/replacement donors on the basis of a higher level of HIV safety. However, recent evidence clearly indicated that, as populations, VNRD and family/replacement donors were not epidemiologically different: the apparent difference being mostly related to VNRD repeating donations following screening. Remarkably few reports have addressed the issue of incidence limiting the ability to predict the residual risks of transfusion-related viral infections and to consider strategies to further reduce such risks (Lefrere *et al.* 2011; Candotti *et al.* 2001).

The present study was made possible by the availability of computerized entries of donors and donations in the three main regional blood services. By studying separately first-time and repeat voluntary donations, both true prevalence and incidence of three infectious diseases were calculated.

VNRD in Burkina Faso are mostly young men (80.8%) who present more often and more often repeat donations than women. As previously indicated in Burkina Faso and other West African countries, the percentage of repeat

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Table 3 Incidence of HIV, HBV and HCV among repeat and first donation in three regional blood centres in Burkina Faso

	HIV			HBV			HCV			HIV	HBV	HCV
	Ouagadougou	Bobo-Dioulasso	Fada†	Ouagadougou	Bobo-Dioulasso	Fada†	Ouagadougou	Bobo-Dioulasso	Fada†	Total	Total	Total
Total donations	22 214	11 180	6512	22 214	11 180	6512	22 214	11 180	6512	39 906	39 906	39 906
Repeat donations												
N (healthy)	4446	2859	1046	4446	2859	1046	4446	2859	1046	8351	8351	8351
N (infected)	36	9	8	75	9	12	73	19	22	53	96	114
IR/100 000 donations	3679.1	1336.5	3340.1	7567.9	1336.4	4986.0	7380.7	2812.4	9080.3	2802.2	5038.9	5978.2
First-time donations												
N (total)	17 292	8246	5379	17 292	8246	5379	17 292	8246	5379	30 917	30 917	30 917
N (infected)	425	81	61	2334	901	968	1116	353	495	567	4203	1964
Conversion factor	3.04	3.12	1.48	8.00	34.71	15.69	3.93	6.44	4.38	2.89	11.83	4.65
IR/100 000 donations	11 167.5	4170.6	4952.6	60 553.3	46 385.7	78 213.0	29 010.9	18 116.2	39 729.6	8097.4	59 589.3	27 819.3

IR, incidence rate.
†Fada N'gourma.

donors is low (11.3%); most of them donate only twice a year (Table 1; Owusu-Ofori *et al.* 2010; Diarra *et al.* 2009). The prevalence of infectious disease markers in first-time VNRD across the country is similar to what has been described in the literature for the general population (Collenberg *et al.* 2006) and considerably higher than data reported from Bobo-Dioulasso (Dahourou *et al.* 2010; Lefrere *et al.* 2011) that are not representative of the whole country. Epidemiologic differences between blood centres are considerable and highly significant (Table 2). The prevalence of anti-HIV and anti-HBV is quite similar to what has been described in neighbouring countries such as Mali, Guinea and Ghana in family replacement donors (Loua & Nze Nkoure 2010; Diarra *et al.* 2009; Allain *et al.* 2008). In contrast, the prevalence of anti-HCV ranging between 4.3% and 9.2% according to blood centres in Burkina Faso was considerably higher than that described in other West African countries (Candotti *et al.* 2003; Jeannel *et al.* 1998; Ruggieri *et al.* 1996). Anti-HCV was found preferentially in younger donors (80.3% below 30 years of age vs. 75.4% <30 in non-HCV-infected donors). Seroconversion to anti-HCV was also found in high frequency in repeat donors (2.75%/year), suggesting that routes of infection remain to be uncovered.

Considering the high frequency of all markers, co-infection with one or several infectious diseases had high prevalence (Table 2). However, the prevalences of co-infections between HIV, HBV and HCV were similar to the single-marker prevalence, suggesting very little overlap of routes of infection. In sub-Saharan Africa, the preferential route of HBV infection for instance is horizontal between young children, explaining the high prevalence of chronic infections, so that by the time they are sexually active and at risk of HIV infection, chronic HBV infection has been established for a decade or more. However, the high frequency of HBV seroconversion in repeat donors (2.5%/year) suggests that sexual transmission takes place in susceptible adults. HCV transmission being associated with direct blood contact, no association between either HIV or HBV infection was observed.

The incidence of all three viral infections was high among the three blood transfusion centres. The prevalence of HIV in first-time VNRD and repeat donors was very close (1.8% and 1.3%), quite distant for HBsAg (13.4% and 2.4%) and closer for anti-HCV (6.3% and 2.9%, respectively). The relatively small difference in HIV and HCV prevalence among first-time and repeat blood donors clearly indicates that blood donor selection does not guarantee high level of safety. First-time volunteers' epidemiology appears fairly representative of the general population and constitutes bulk of the blood supply. As a

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result, only retention of repeat donors is able to improve blood safety in the three main blood centres in Burkina Faso. In addition, considering the chronic blood shortage in the country and the high cost of blood collected from volunteer donors, the reintroduction of blood collection from family donors should be seriously examined in a resource-limited country drawing on its own healthcare budget to supply blood and components to a majority of patients whose survival depends on blood availability.

Conclusion

The relatively high prevalence of viral markers in first-time volunteers and remarkably high incidence of infections in repeat donors raise concerns regarding the safety of these donors and suggest that implementation of NAT might significantly improve the situation.

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